

Marie Bruneau

Clinical genetics practice · AI-assisted tool development · Clinician-facing applications
for documentation, education and research workflows

France | +33 6 76 64 29 75 | Marie.Bruneau@outlook.be | ORCID: 0000-0003-3746-5752 | GitHub: MKB07

PROFIL

EBMG-registered Genetic Counsellor with 7 years of clinical experience in human genetics and a self-directed software-development profile. Focused on translating clinical expertise into practical Health AI tools, combining R/ Shiny, pedtools, trusted biomedical data sources, human-centred design and responsible generative AI prototyping.

CORE CLINICAL SKILLS

- Genetic counselling, risk assessment & pedigree analysis
- Adult, paediatric, oncogenetics, cardiogenetics
- Prenatal / preconception genetics
- Clinical documentation & multidisciplinary meetings
- Polygenic risk scores, CanRisk / BOADICEA

TECHNICAL TOOLS

- R / Shiny (self-directed learning)
- Basic Python, HTML/CSS and C exposure
- APIs & biomedical databases
- Excel, REDCap, Access, GLIMS (MIPS)
- VCF, SNP-array / imputation concepts & QC
- Generative AI prototyping (learning)

CREDENTIALS

- EBMG — Registered Genetic Counsellor (EU)
- Human Genetics & Medicine/AI certificates (ULB)
- OpenAI Academy: AI Foundations, Agents & Workflows (2026)
- Google ML Crash Course & Learning-profile badges (2026)

LANGUAGES

French Native

English Professional proficiency

Spanish Basic knowledge

VOLUNTEERING

- Athletics coach for young athletes
- AIDES association — awareness campaigns and support for people living with HIV

EDUCATION

Certificate in Medicine & Artificial Intelligence Université libre de Bruxelles (ULB)	11/2025 — Present
Certificate in Human Genetics Université libre de Bruxelles (ULB)	10/2021 — 01/2022
Master's in Genetic Counselling & Predictive Medicine Aix-Marseille University, Faculty of Medicine	10/2015 — 10/2017
Bachelor's in Animal Physiology & Neuroscience University of Poitiers, Faculty of Fundamental & Applied Sciences	09/2012 — 06/2015

PROFESSIONAL EXPERIENCE

Independent Genetic Counsellor UNIQUE · Switzerland · contract-based clinical genetics consultancy	Current
- Prepared clinical genetics consultations and patient files in the context of independent consultancy work. - Collected and analysed personal and family histories; established and updated pedigrees. - Coordinated genetic analyses and contributed to draft reports and professional correspondence. - Participated in genetic counselling consultations and prepared information materials for patients. - Contributed to genetics research projects, scientific publications and presentations; managed patient registries when required.	

Genetic Counsellor ULB Centre of Human Genetics · Brussels, Belgium	02/2018 — 11/2024
- Conducted genetic counselling consultations in adult, paediatric, oncogenetics, cardiogenetics, preconception and prenatal settings — risk assessment, family-history analysis and patient-centred guidance. - Managed and followed patient records, ensuring accurate clinical updates to support medical decision-making and continuity of care. - Prepared and coordinated multidisciplinary meetings, synthesising clinical and genetic information. - Contributed to polygenic risk scoring research — data analysis, interpretation and clinical reflection. - Provided GLIMS software support: user training, troubleshooting and workflow optimisation. - Supervised and trained genetic counselling interns.	

SELECTED PROJECTS

AI AI-Assisted Clinical Genetics Platform Genetic Pedigree Builder Web application supporting pedigree creation, family-history exploration, clinical genetics education and research workflows. Built with R/ Shiny and AI-assisted coding. Next prototype explores generative AI for draft consultation reports — demonstration only, not for clinical decision-making. R Shiny pedtools APIs ChatGPT/Codex	PRS Breast Cancer PRS Feasibility Project ULB Centre · BeSHG Presentation Evaluated polygenic risk score production in a routine clinical genetics laboratory. Explored PRS interaction with personal, familial and clinical risk factors. Identified priorities for automated imputation, QC reporting and clinical counselling integration. CanRisk/BOADICEA BACC313 PRS SNP-array VCF
--	--

CLINICAL INTERNSHIPS

McGill University Health Centre & Jewish General Hospital Montréal, Canada — Paediatric and adult medical genetics; oncogenetics.	07-09/2017
University Hospitals of Geneva Geneva, Switzerland — Oncogenetics; additional experience in medical genetics.	04-05/2017
Necker-Enfants Malades Hospital Paris, France — Prenatal diagnosis; specialisation in sickle cell disease.	06-07/2016

PUBLICATIONS

- Neunes, Sebastien et al. "Diagnostic yield of clinical exome sequencing in 868 children with neurodevelopmental disorders." **European Journal of Medical Genetics** vol. 76 (2025): 105030. doi: 10.1016/j.ejmg.2025.105030
- Castiaux, Julie et al. "Case Report of an Unusual Tumor in an Adult With a TP53 Germline Mutation." **Clinical Lymphoma, Myeloma & Leukemia** vol. 21,8 (2021): e645-e648. doi: 10.1016/j.clml.2020.10.001